

Rates of Change and Rectilinear Motion



Position, velocity, and acceleration

- 1 A particle moves along a line with position $s(t) = t^3 - 6t^2 + 9t$ (metres, seconds).
 - a Find the velocity function $v(t)$.
 - b Find the acceleration function $a(t)$.
 - c Find $v(2)$ and $a(2)$.
 - 2 For the particle above, find all times $t \geq 0$ when the particle is at rest ($v(t) = 0$).
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Speeding up and slowing down

- 3 Using the particle from above, determine whether the particle is speeding up or slowing down at $t = 2$ (compare the signs of $v(2)$ and $a(2)$).
 - 4 Determine whether the particle is speeding up or slowing down at $t = 0$.
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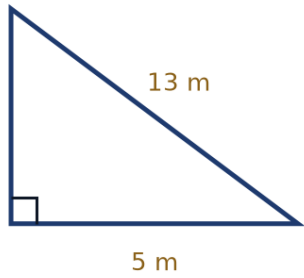
Total distance vs. displacement

- 5 For the particle above, find its displacement over $[0, 3]$.
 - 6 Using the fact that the particle changes direction at $t = 1$ (found earlier), find the total distance traveled over $[0, 3]$.
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Related rates

- 7 A spherical balloon is inflated so its radius grows at 2 cm/s. Find the rate of change of its volume when $r = 6$ cm, using $V = \frac{4}{3}\pi r^3$.

- 8 A 13 m ladder leans against a wall. The base slides away from the wall at 3 m/s. Find how fast the top of the ladder is sliding down when the base is 5 m from the wall.

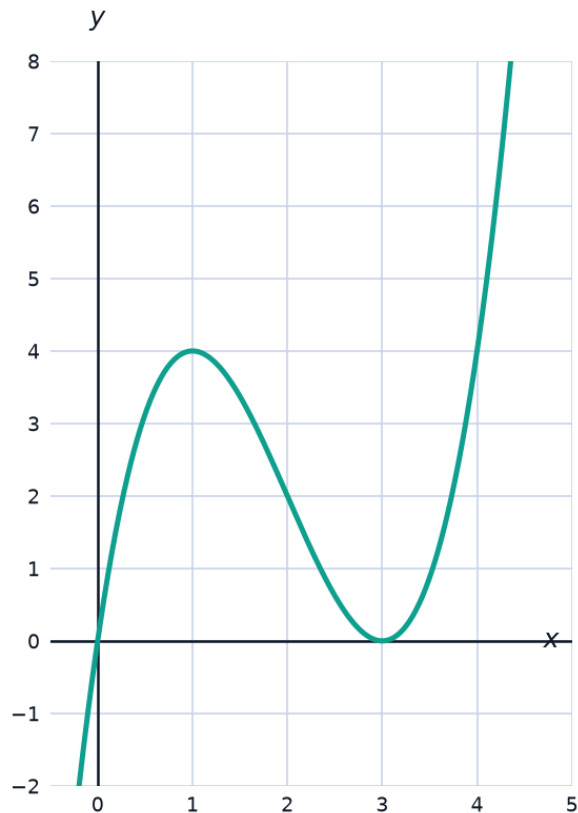


Motion from a velocity graph

- 9 A particle's velocity is $v(t) = t^2 - 4t$ (m/s) for $t \geq 0$. Find the times when the particle changes direction.
- 10 For the same particle, find the acceleration at $t = 1$ and state whether the particle is speeding up or slowing down at that instant.

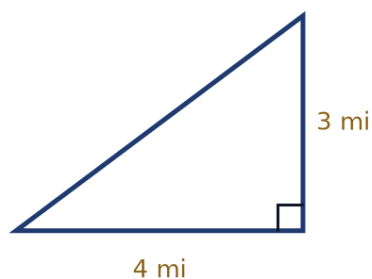
Visualizing position, velocity, and acceleration together

- 11 The graph shows the position function $s(t) = t^3 - 6t^2 + 9t$ from the first problem. Use the graph to identify the interval(s) where the particle is moving to the left (i.e. s is decreasing).



An applied related-rates capstone

- 12 A plane flies at a constant altitude of 3 miles, moving horizontally at 500 mph, directly away from a radar station on the ground. Find the rate at which the distance between the plane and the radar station is increasing when the plane is 4 miles horizontally from the station (directly above a point 4 miles from the station).



Motion with a velocity given as a formula

- 13 A particle's velocity is $v(t) = 6t - t^2$ (m/s) for $0 \leq t \leq 8$. Find the particle's position function $s(t)$ given $s(0) = 0$, then find its displacement over $[0, 6]$.
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A capstone rectilinear motion problem

- 14 A rocket's height is $h(t) = -5t^2 + 40t$ (meters, seconds) after launch. Find its maximum height, and the time at which it returns to the ground.
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Multiple choice — easy 15

- 15 A particle's position is $s(t) = -5t^2 + 2t - 4$ (meters, seconds). Find its velocity $v(4)$.
- a. -18
 - b. -36
 - c. -40
 - d. -38
- 16 A particle's position is $s(t) = -6t^2 - 3t + 9$ (meters, seconds). Find its velocity $v(5)$.
- a. -65
 - b. -63
 - c. -61
 - d. -33
- 17 A particle's position is $s(t) = 6t^2 + 9t - 5$ (meters, seconds). Find its velocity $v(3)$.
- a. 27
 - b. 45
 - c. 47
 - d. 43

- 18 A particle's position is $s(t) = -2t^2 + t + 8$ (meters, seconds). Find its velocity $v(4)$.
- a. -13
 - b. -17
 - c. -7
 - d. -15
- 19 A particle's position is $s(t) = t^2 - 5t - 9$ (meters, seconds). Find its velocity $v(3)$.
- a. -2
 - b. -1
 - c. 3
 - d. 1
- 20 A particle's position is $s(t) = -6t^2 + 7t - 4$ (meters, seconds). Find its velocity $v(0)$.
- a. 9
 - b. 8
 - c. 7
 - d. 5
- 21 A particle's position is $s(t) = 3t^2 - 8t + 3$ (meters, seconds). Find its velocity $v(4)$.
- a. 14
 - b. 4
 - c. 16
 - d. 18

- 22 A particle's position is $s(t) = 4t^2 - 5t$ (meters, seconds). Find its velocity $v(4)$.
- a. 11
 - b. 25
 - c. 27
 - d. 29
- 23 A particle's position is $s(t) = -6t^2 + 7t - 9$ (meters, seconds). Find its velocity $v(3)$.
- a. -27
 - b. -31
 - c. -29
 - d. -11
- 24 A particle's position is $s(t) = -5t^2 - 3t + 6$ (meters, seconds). Find its velocity $v(2)$.
- a. -23
 - b. -25
 - c. -13
 - d. -21
- 25 A particle's position is $s(t) = t^2 + 9t + 4$ (meters, seconds). Find its velocity $v(5)$.
- a. 19
 - b. 14
 - c. 17
 - d. 21

- 26 A particle's position is $s(t) = 4t^2 + 9t$ (meters, seconds). Find its velocity $v(2)$.
- a. 25
 - b. 23
 - c. 27
 - d. 17
- 27 A particle's position is $s(t) = 5t^2 + 5t - 7$ (meters, seconds). Find its velocity $v(3)$.
- a. 35
 - b. 20
 - c. 37
 - d. 33
- 28 A particle's position is $s(t) = 3t^2 - 4t - 8$ (meters, seconds). Find its velocity $v(1)$.
- a. 4
 - b. -1
 - c. 2
 - d. 0
- 29 A particle's position is $s(t) = -5t^2 + 6t - 8$ (meters, seconds). Find its velocity $v(1)$.
- a. -4
 - b. 1
 - c. -2
 - d. -6

Multiple choice — medium 15

- 30 A particle's position is $s(t) = t^3 + 6t$ (meters, seconds). Find its acceleration $a(3)$.
- a. 12
 - b. 18
 - c. 24
 - d. 9
- 31 A particle's position is $s(t) = t^3 + t^2 - 2t$ (meters, seconds). Find its acceleration $a(2)$.
- a. 20
 - b. 14
 - c. 7
 - d. 8
- 32 A particle's position is $s(t) = t^3 - 4t^2 + 3t$ (meters, seconds). Find its acceleration $a(1)$.
- a. -1
 - b. -2
 - c. 4
 - d. -8
- 33 A particle's position is $s(t) = t^3 - 6t^2 - 8t$ (meters, seconds). Find its acceleration $a(5)$.
- a. 24
 - b. 9
 - c. 12
 - d. 18

- 34 A particle's position is $s(t) = t^3 - 2t^2 + 8t$ (meters, seconds). Find its acceleration $a(3)$.
- a. 20
 - b. 8
 - c. 7
 - d. 14
- 35 A particle's position is $s(t) = t^3 - 2t^2 - 2t$ (meters, seconds). Find its acceleration $a(2)$.
- a. 8
 - b. 4
 - c. 14
 - d. 2
- 36 A particle's position is $s(t) = t^3 - 4t^2 - 5t$ (meters, seconds). Find its acceleration $a(3)$.
- a. 5
 - b. 4
 - c. 16
 - d. 10
- 37 A particle's position is $s(t) = t^3 + t^2 - 7t$ (meters, seconds). Find its acceleration $a(1)$.
- a. 2
 - b. 4
 - c. 14
 - d. 8

- 38 A particle's position is $s(t) = t^3 + 6t^2 + 6t$ (meters, seconds). Find its acceleration $a(4)$.
- a. 36
 - b. 18
 - c. 30
 - d. 42
- 39 A particle's position is $s(t) = t^3 - 4t$ (meters, seconds). Find its acceleration $a(0)$.
- a. 5
 - b. -6
 - c. 0
 - d. 6
- 40 A particle's position is $s(t) = t^3 - t^2 - 5t$ (meters, seconds). Find its acceleration $a(5)$.
- a. 22
 - b. 34
 - c. 28
 - d. 14
- 41 A particle's position is $s(t) = t^3 + t^2 + 6t$ (meters, seconds). Find its acceleration $a(2)$.
- a. 20
 - b. 8
 - c. 14
 - d. 7

- 42 A particle's position is $s(t) = t^3 + 5t^2 - t$ (meters, seconds). Find its acceleration $a(2)$.
- a. 28
 - b. 22
 - c. 16
 - d. 11
- 43 A particle's position is $s(t) = t^3 + 6t^2 - t$ (meters, seconds). Find its acceleration $a(1)$.
- a. 24
 - b. 18
 - c. 12
 - d. 9
- 44 A particle's position is $s(t) = t^3 + 5t^2 + 2t$ (meters, seconds). Find its acceleration $a(3)$.
- a. 28
 - b. 14
 - c. 22
 - d. 34

Multiple choice — hard 15

- 45 A particle's velocity is $v(t) = t^2 - t + 4$ (m/s). Determine whether the particle is speeding up or slowing down at $t = 2$.
- a. $v(2) = 6$ and $a(2) = 3$ — same sign, so the particle is slowing down.
 - b. $v(2) = 6$ and $a(2) = 3$ — opposite signs, so the particle is speeding up.
 - c. $v(2) = 6$ and $a(2) = 3$ — same sign, so the particle is speeding up.
 - d. $v(2) = 6$ and $a(2) = 3$ — opposite signs, so the particle is slowing down.

- 46 A particle's velocity is $v(t) = t^2 + 5t - 4$ (m/s). Determine whether the particle is speeding up or slowing down at $t = 2$.
- a. $v(2) = 10$ and $a(2) = 9$ — same sign, so the particle is slowing down.
 - b. $v(2) = 10$ and $a(2) = 9$ — same sign, so the particle is speeding up.
 - c. $v(2) = 10$ and $a(2) = 9$ — opposite signs, so the particle is slowing down.
 - d. $v(2) = 10$ and $a(2) = 9$ — opposite signs, so the particle is speeding up.
- 47 A particle's velocity is $v(t) = t^2 + 4t + 2$ (m/s). Determine whether the particle is speeding up or slowing down at $t = 3$.
- a. $v(3) = 23$ and $a(3) = 10$ — same sign, so the particle is speeding up.
 - b. $v(3) = 23$ and $a(3) = 10$ — opposite signs, so the particle is speeding up.
 - c. $v(3) = 23$ and $a(3) = 10$ — same sign, so the particle is slowing down.
 - d. $v(3) = 23$ and $a(3) = 10$ — opposite signs, so the particle is slowing down.
- 48 A particle's velocity is $v(t) = t^2 - 2t + 3$ (m/s). Determine whether the particle is speeding up or slowing down at $t = 0$.
- a. $v(0) = 3$ and $a(0) = -2$ — opposite signs, so the particle is speeding up.
 - b. $v(0) = 3$ and $a(0) = -2$ — opposite signs, so the particle is slowing down.
 - c. $v(0) = 3$ and $a(0) = -2$ — same sign, so the particle is slowing down.
 - d. $v(0) = 3$ and $a(0) = -2$ — same sign, so the particle is speeding up.
- 49 A particle's velocity is $v(t) = t^2 - 4t + 6$ (m/s). Determine whether the particle is speeding up or slowing down at $t = 1$.
- a. $v(1) = 3$ and $a(1) = -2$ — opposite signs, so the particle is speeding up.
 - b. $v(1) = 3$ and $a(1) = -2$ — same sign, so the particle is slowing down.
 - c. $v(1) = 3$ and $a(1) = -2$ — opposite signs, so the particle is slowing down.
 - d. $v(1) = 3$ and $a(1) = -2$ — same sign, so the particle is speeding up.

- 50 A 17 m ladder leans against a wall. The base slides away from the wall at 1 m/s. Find how fast the top is sliding down when the base is 8 m from the wall.
- a. -1
 - b. $-\frac{8}{15}$
 - c. $\frac{15}{8}$
 - d. $\frac{8}{15}$
- 51 A 10 m ladder leans against a wall. The base slides away from the wall at 5 m/s. Find how fast the top is sliding down when the base is 6 m from the wall.
- a. $\frac{20}{3}$
 - b. -5
 - c. $-\frac{15}{4}$
 - d. $\frac{15}{4}$
- 52 A 5 m ladder leans against a wall. The base slides away from the wall at 4 m/s. Find how fast the top is sliding down when the base is 3 m from the wall.
- a. $\frac{16}{3}$
 - b. -3
 - c. -4
 - d. 3
- 53 A 15 m ladder leans against a wall. The base slides away from the wall at 3 m/s. Find how fast the top is sliding down when the base is 9 m from the wall.
- a. 4
 - b. -3
 - c. $-\frac{9}{4}$
 - d. $\frac{9}{4}$

- 54 A 13 m ladder leans against a wall. The base slides away from the wall at 1 m/s. Find how fast the top is sliding down when the base is 5 m from the wall.
- a. -1
 - b. $\frac{12}{5}$
 - c. $-\frac{5}{12}$
 - d. $\frac{5}{12}$
- 55 A particle's position is $s(t) = t^3 - 12t^2 + 45t$ (meters, seconds), which is at rest at $t = 3$ and $t = 5$. Find the total distance traveled over $[0, 5]$.
- a. 50
 - b. 60
 - c. 58
 - d. 56
- 56 A particle's position is $s(t) = t^3 - 9t^2 + 15t$ (meters, seconds), which is at rest at $t = 1$ and $t = 5$. Find the total distance traveled over $[0, 5]$.
- a. 25
 - b. 37
 - c. 39
 - d. 41
- 57 A particle's position is $s(t) = t^3 - 15t^2 + 72t$ (meters, seconds), which is at rest at $t = 4$ and $t = 6$. Find the total distance traveled over $[0, 6]$.
- a. 116
 - b. 114
 - c. 108
 - d. 118

- 58 A particle's position is $s(t) = t^3 - 9t^2 + 24t$ (meters, seconds), which is at rest at $t = 2$ and $t = 4$. Find the total distance traveled over $[0, 4]$.
- a. 16
 - b. 24
 - c. 26
 - d. 22
- 59 A particle's position is $s(t) = t^3 - 15t^2 + 63t$ (meters, seconds), which is at rest at $t = 3$ and $t = 7$. Find the total distance traveled over $[0, 7]$.
- a. 111
 - b. 113
 - c. 115
 - d. 49